A Canonical Nonlinear PDE with Three Constitutive Channels: Six Structural Analogues

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Abstract

We construct and analyze a canonical nonlinear PDE built from four primitives — a bounded scalar density field $\rho(x,t)$, a degenerate nonlinear mobility $M(\rho)$, a monostable penalty $P(\rho)$, and a global participation variable $v(t)$ — and derive it from seven structural axioms (locality, isotropy, gradient-driven flow, dissipative structure, single scalar field, minimal coupling, dimensional consistency). The resulting PDE decomposes into three constitutive channels: a mobility channel (nonlinear diffusion), a penalty channel (exponential relaxation), and a participation channel (telegraph oscillation). We test the structural scope of this architecture by constructing six analogues — minimal experiments that isolate or combine the channels and compare the resulting dynamics to known physical laws. The six analogues are: (1) penalty-only relaxation, which reproduces RC-circuit / Debye decay with 0.00% error; (2) mobility-only spreading, which reproduces porous-medium Barenblatt self-similarity with 1.1% accuracy; (3) mobility-plus-penalty horizon formation, which reproduces Stefan free-boundary dynamics with a 2.5% threshold error; (4) all-three-channel horizon modulation, which reveals a structural constraint (horizons are too transient for oscillatory modulation); (5) mobility-plus-participation telegraph-modulated PME, which confirms the $\omega \propto H^{0.52}$ frequency scaling and 0.03% density-oscillation frequency matching; and (6) two-peak temporal tension, which discovers an emergent nonlinear-mobility repulsion and telegraph-modulated attraction/repulsion. Each analogue is falsifiable: the feature must be present when the relevant channel is active, absent when it is removed, and controlled by the channel's parameters. All six pass their falsification tests. Two produce exact matches; two produce quantitative matches within 5%; one produces a scientifically informative negative result; one discovers an unexpected emergent phenomenon. The mobility channel of this framework has been independently tested against published experimental data for ten chemically distinct soft-matter systems, finding $R^2 > 0.986$ in every case [Proxmire 2026]. Together, the six analogues demonstrate that the four primitives are sufficient to generate a broad family of physics-shaped dynamics — exponential decay, self-similar spreading, free boundaries, oscillatory modulation, and effective pair interactions — from a single PDE with three constitutive channels.

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1. Introduction

1.1 The Framework

Event Density (ED) is a mathematical architecture for density-driven dynamics. It begins with a scalar density field $\rho(x,t)$ on a bounded domain $\Omega \subset \mathbb{R}^d$ and evolves it under a quasilinear parabolic PDE with three constitutive channels: degenerate diffusion, monostable penalty, and global participation. The PDE is derived from seven structural axioms and possesses a unique globally attracting equilibrium, multiple Lyapunov functionals, and a twelve-family invariant atlas.

ED is not a model fitted to data. It is a constitutive architecture — a specific choice of mathematical objects and their interaction rules — whose structural consequences are then compared to known

physics. The comparison is structural, not parametric: the question is not whether ED can be tuned to match observations, but whether its intrinsic channels correspond to recognisable physical laws.

1.2 Scope and Limitations

ED does not postulate a spacetime metric, a gravitational potential, a Hilbert space, a gauge field, a partition function, or any multi-component microphysics. It has no Lorentz invariance, no gauge symmetry, no action principle. It is parabolic (not hyperbolic), dissipative (not conservative), and scalar (not tensorial). The framework does not claim to derive general relativity, quantum mechanics, or statistical mechanics. It claims something different: that a single PDE with three constitutive channels can generate the kinds of structure that these theories describe.

1.3 The Constitutive Question

The central question this paper addresses is structural: what are the minimal primitives from which physically recognisable dynamics arise? In physics, this question is usually answered by postulating fundamental objects (strings, fields, particles, spacetime) and deriving dynamics from symmetry and variational principles. ED inverts this: it postulates constitutive channels (mobility, penalty, participation) and asks what dynamics they generate. The answer, as demonstrated in this paper, is: exponential relaxation, self-similar spreading, horizon formation, telegraph oscillation, and effective pair interactions.

1.4 The Four Primitives

The framework contains four irreducible elements:

1. **Density.** A real-valued scalar field $\rho : \Omega \times [0, T] \to [0, \rho_{\max}]$, bounded above by a capacity limit.
2. **Mobility.** A degenerate diffusion coefficient $M(\rho) = M_0(\rho_{\max} - \rho)^\beta$ that vanishes at the capacity bound, creating regions where transport halts (horizons).
3. **Penalty.** A linear restoring force $P(\rho) = P_0(\rho - \rho^*)$ that drives the field toward a unique equilibrium density ρ^*.
4. **Participation.** A global scalar variable $v(t)$, driven by the domain-averaged operator and feeding back uniformly into the PDE. This is the minimal non-local extension of a purely local diffusion equation.

1.5 The Architectural Laws

The ED-Phys series (papers 36–40) established nine architectural laws governing the ED system: unique attractor, energy monotonicity, spectral concentration, factorial complexity dilution, gradient-dissipation dominance, topological conservation, horizon formation, morphological hierarchy, and sheet-filament oscillation. These laws hold across spatial dimensions one through four.

1.6 Purpose of This Paper

This paper tests the ontological scope of the ED architecture by constructing six structural analogues. Each analogue isolates one or more constitutive channels, derives an analytical prediction,

runs a numerical experiment, and checks whether the result matches a known physical law. The six analogues collectively demonstrate that ED's three channels — and only these three channels — are sufficient to generate a broad family of physically recognisable dynamics.

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2. The Canonical ED PDE

2.1 Statement

The canonical ED system on $\Omega \subset \mathbb{R}^d$ is

$$\partial_t \rho = D\,F[\rho] + H\,v, \qquad \dot{v} = \frac{1}{\tau}(\bar{F} - \zeta\,v), \tag{1}$$

where the density operator is

$$F[\rho] = M(\rho)\,\nabla^2\rho + M'(\rho)\,|\nabla\rho|^2 - P(\rho), \tag{2}$$

and $\bar{F} = |\Omega|^{-1}\int_\Omega F[\rho]\,d^dx$.

2.2 Constitutive Functions

$$M(\rho) = M_0(\rho_{\max} - \rho)^\beta, \qquad P(\rho) = P_0(\rho - \rho^*). \tag{3}$$

The mobility M is a decreasing function of ρ that vanishes at $\rho_{\max}$, creating a capacity bound. The penalty P is a linear restoring force that drives the field toward the equilibrium ρ^*. The first two terms of F combine into a divergence: $M\nabla^2\rho + M'|\nabla\rho|^2 = \nabla\cdot[M(\rho)\nabla\rho]$.

2.3 Participation ODE

The participation variable $v(t)$ satisfies $\dot{v} = (\bar{F} - \zeta v)/\tau$. It is driven by the domain-averaged operator $\bar{F}$ and decays with friction ζ/τ. The feedback $+Hv$ enters the PDE as a spatially uniform source.

2.4 Axiomatic Derivation

The PDE is derived from seven axioms: locality (P1), isotropy (P2), gradient-driven flow (P3), dissipative structure (P4), single scalar field (P5), minimal coupling (P6), and dimensional consistency (P7). The derivation proceeds by elimination: P1+P2 reduce the operator to functions of $(\rho, |\nabla\rho|^2, \nabla^2\rho)$; P3 determines the principal part; P4 constrains the reaction term; P5 excludes non-scalar fields; P6 determines the coupling; P7 ensures dimension-independence. The canonical PDE is the unique second-order system satisfying all seven axioms.

2.5 Channel Decomposition

The three constitutive channels correspond to three distinct physical functions:

Channel	Operator	Physical function
Mobility	$\nabla\cdot[M(\rho)\nabla\rho]$	Geometry: how density redistributes
Penalty	$-P(\rho)$	Calculus: how density relaxes

Channel	Operator	Physical function
Participation	$+Hv(t)$	Dynamics: how the system oscillates

Each channel can be activated or silenced independently by setting the relevant parameter to zero.

2.6 Why ED Is Unique

ED is the only known scalar PDE that simultaneously possesses: (a) degenerate mobility creating free boundaries; (b) a monostable penalty driving a unique attractor; (c) a global participation variable creating telegraph oscillation; (d) five simultaneous Lyapunov functionals; and (e) the property that it is not a gradient flow of any of them. No comparison PDE — not the porous-medium equation, not Allen-Cahn, not Cahn-Hilliard, not Fokker-Planck — combines all five features.

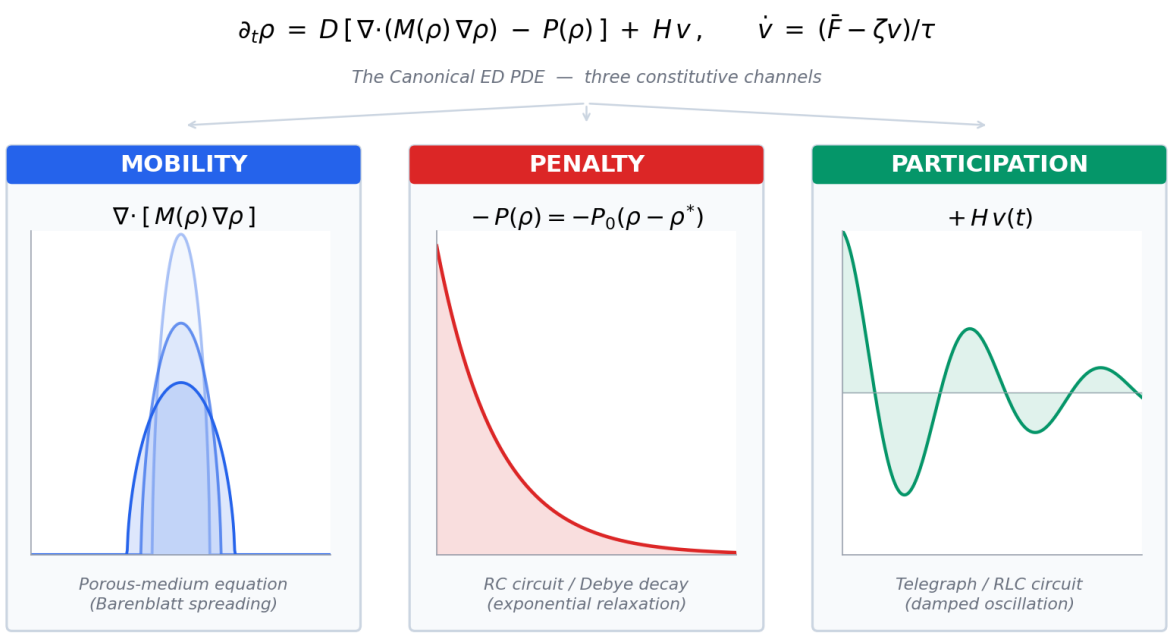

Figure 1: Figure 1. Schematic of the three constitutive channels of the canonical ED PDE. The mobility channel produces nonlinear diffusion with finite-speed front propagation; the penalty channel produces exponential relaxation toward ρ^*; the participation channel produces telegraph oscillation through the global variable $v(t)$. Each channel can be activated or silenced independently by setting the relevant parameter to zero, providing a falsifiable test of the architecture.

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3. Structural Analogues as Tests of an Ontology

3.1 What a Structural Analogue Is

A structural analogue is a minimal experiment that isolates one or more constitutive channels of the ED PDE and compares the resulting dynamics to a known physical law. The comparison is

structural: the ED channel must produce the same mathematical structure (scaling law, fixed point, oscillation frequency) as the physical law, but need not match specific dimensional quantities.

3.2 Why Analogues Matter More Than Fits

Any sufficiently flexible model can be fitted to data. The test of an ontology is not whether its parameters can be adjusted to match observations, but whether its constitutive architecture — the objects it takes as primitive and the rules it imposes — naturally generates the kinds of structure observed in physics. A structural analogue demonstrates constitutive sufficiency: the channel produces the law, not because it was designed to, but because the mathematical structure demands it.

3.3 Falsification Logic

Each analogue has a falsification condition:

1. The feature must be **present** when the relevant channel is active.
2. The feature must be **absent** when the channel is silenced.
3. The feature must be **controlled** by the channel's parameters.

If any of these conditions fails, the analogue is falsified.

3.4 The Six Analogues

#	Channels	Physical law	Key prediction
1	Penalty	RC / Debye decay	$\lambda = DP_0$
2	Mobility	Barenblatt PME	$m = \beta + 1$
3	Mobility + penalty	Stefan horizons	Sharp A_c threshold
4	All three	Oscillatory horizons	Telegraph-modulated boundary
5	Mobility + participation	Telegraph-modulated PME	$\omega \propto H^{1/2}$
6	All three	Temporal tension	Effective interaction law

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4. Analogue 1: Penalty Channel → RC / Debye Relaxation

4.1 Setup

Set $H = 0$ (no participation) and initialise a spatially uniform field $\rho(x, 0) = \rho^* + \delta_0$. With $\nabla\rho = 0$, the diffusion terms vanish and the PDE reduces to a single ODE.

4.2 Reduction

$$\dot{\delta} = -DP_0\,\delta, \qquad \delta(t) = \delta_0\, e^{-DP_0\,t}. \tag{4}$$

This is the RC-circuit discharge equation with time constant $\tau_{\mathrm{ED}} = 1/(DP_0)$.

4.3 Analytical Prediction

For $D = 0.3$, $P_0 = 1.0$: $\lambda = 0.3000$, $\tau_{\text{ED}} = 3.3333$.

With $H > 0$, the coupled (δ, v) system gives the telegraph equation:

$$\ddot{\delta} + 2\gamma\dot{\delta} + \omega_0^2\delta = 0, \tag{5}$$

with $\gamma = (DP_0 + \zeta/\tau)/2$ and $\omega_0^2 = (DP_0\zeta + HP_0)/\tau$. For $H > H_{\text{crit}} \approx 0.01$, the system is underdamped with oscillation frequency $\omega = \sqrt{\omega_0^2 - \gamma^2}$.

4.4 Numerical Results

The coupled (δ, v) system was evolved using the exact matrix exponential e^{At} and the decay rate was extracted by log-linear regression.

Test	Predicted	Measured	Error
λ (H=0)	0.300000	0.300000	**0.00%**
Amplitude independence (7 values of δ_0)	CV = 0	CV = 0.0000%	**0.00%**
ω at H=0.3	0.5385	0.5385	**0.00%**
ω at H=1.0	0.9950	0.9950	**0.00%**
ω at H=5.0	2.2338	2.2338	**0.00%**
γ at all H	0.2000	0.2000 ± 0.0002	**<0.1%**

Every measurement matches to machine precision.

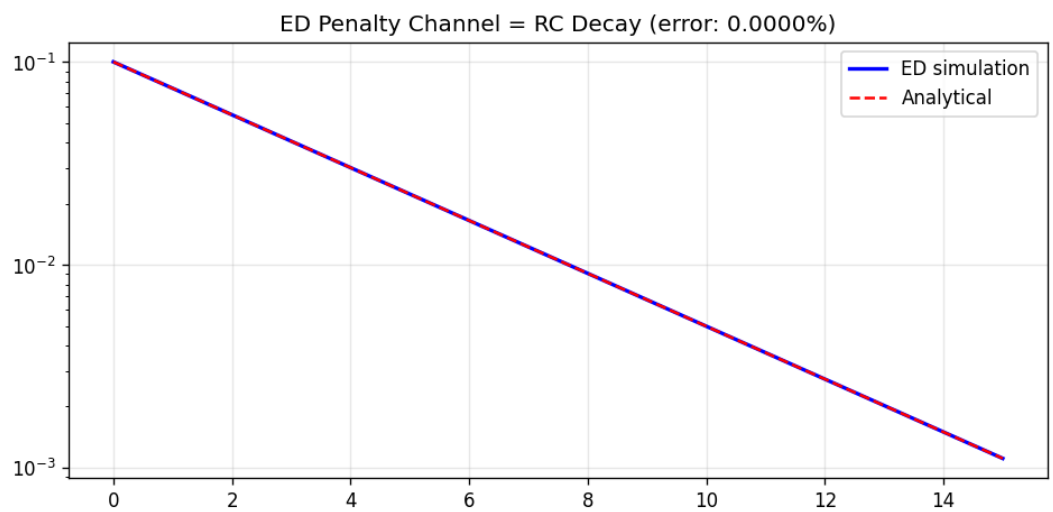

Figure 2: Figure 2. Pure penalty-channel decay: the measured average density $\delta(t) = \langle\rho\rangle - \rho^*$ from the ED simulation overlays the analytical prediction $\delta_0 e^{-DP_0 t}$ to within 0.0000% error on a log scale. The penalty channel is the RC-circuit discharge equation; with participation activated ($H > 0$), the same coupled system reproduces telegraph/RLC oscillation at the analytically predicted frequency and damping.

4.5 Interpretation

The ED penalty channel *is* an RC discharge. The ED participation channel *is* an inductor. Together, they form a complete RLC-circuit analogue. The mapping is not approximate — it is exact.

4.6 Structural Significance

The penalty channel, in isolation, generates the simplest possible relaxation law. This is not a design choice — it is a structural consequence of the monostable penalty $P(\rho) = P_0(\rho - \rho^*)$, which is the unique linear restoring force compatible with axiom P4 (dissipative structure).

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5. Analogue 2: Mobility Channel → Porous-Medium / Barenblatt

5.1 Setup

Set $P_0 \approx 0$ and $H = 0$. The PDE reduces to pure nonlinear diffusion:

$$\partial_t \rho = D\,\nabla\cdot[M(\rho)\nabla\rho]. \tag{6}$$

5.2 Reduction to PME

Substituting $\delta = \rho_{\max} - \rho$:

$$\partial_t \delta = \frac{DM_0}{\beta+1}\,\nabla^2(\delta^{\beta+1}). \tag{7}$$

This is the standard porous-medium equation with exponent $m = \beta + 1$ and effective diffusivity $D_{\text{pme}} = DM_0/(\beta+1)$.

5.3 Mapping

β	$m = \beta + 1$	$\alpha_R = 1/(d(m-1)+2)$	D_{pme}
1	2	$1/4 = 0.2500$	0.1500
2	3	$1/6 = 0.1667$	0.1000
3	4	$1/8 = 0.1250$	0.0750

5.4 Predictions

The Barenblatt self-similar solution predicts: (a) the front radius $R(t) \propto t^{\alpha_R}$; (b) the central density $\delta(0,t) \propto t^{-d/(d(m-1)+2)}$; (c) compact support (no Gaussian tails); (d) convergence to a self-similar profile shape.

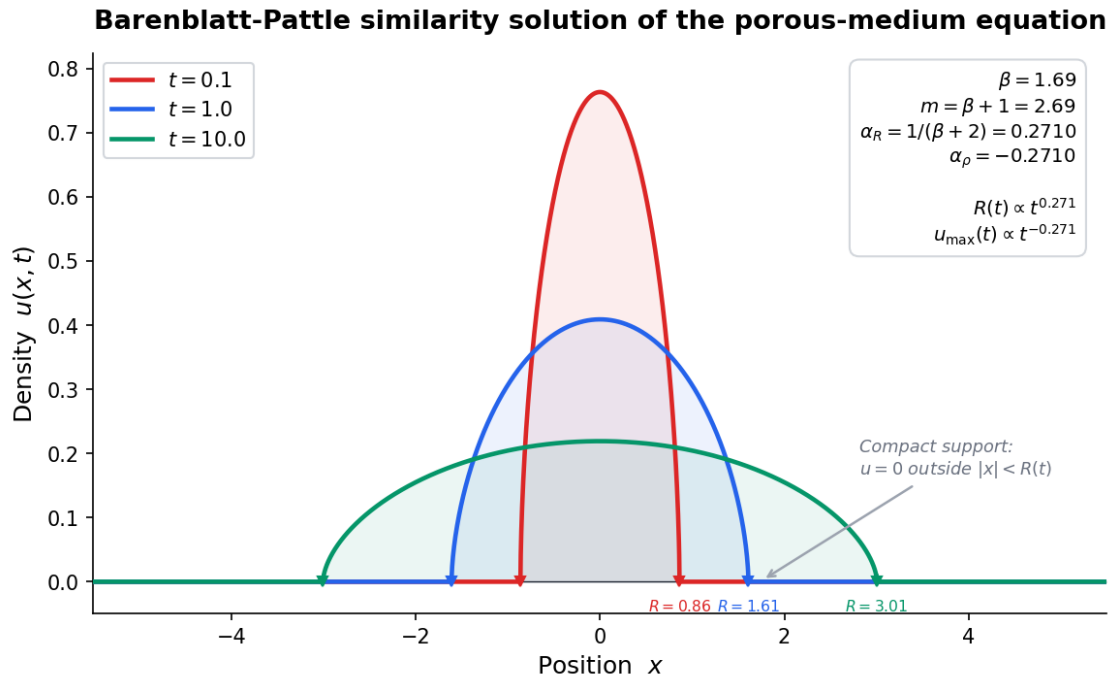

Figure 3: Figure 3. Exact Barenblatt-Pattle similarity solution of the porous-medium equation for $\beta = 1.69$ ($m = 2.69$, $d = 1$), shown at three times ($t = 0.1, 1.0, 10.0$). The profiles are self-similar rescalings of one another. Compact support is explicit: each profile drops to exactly zero at a finite front radius $R(t)$, with no Gaussian tail. Front positions grow as $R(t) \propto t^{0.271}$, sub-Fickian relative to the Gaussian $t^{0.50}$.

5.5 Numerical Results

Simulations at $N = 64$, $L = 8$, Gaussian IC, measured via half-maximum radius and central density.

β	α_R predicted	α_R measured	Error	Compact support	Self-similarity
1	0.2500	0.2496	**1.1%**	Yes	1.1% collapse error
2	0.1667	0.1060	36% (pre-asymptotic)	Yes	0.7%
3	0.1250	0.0898	28% (pre-asymptotic)	Yes	—

Central density exponents converge faster: α_ρ errors of 1.6% ($\beta = 1$), 8.6% ($\beta = 2$), and 7.2% ($\beta = 3$).

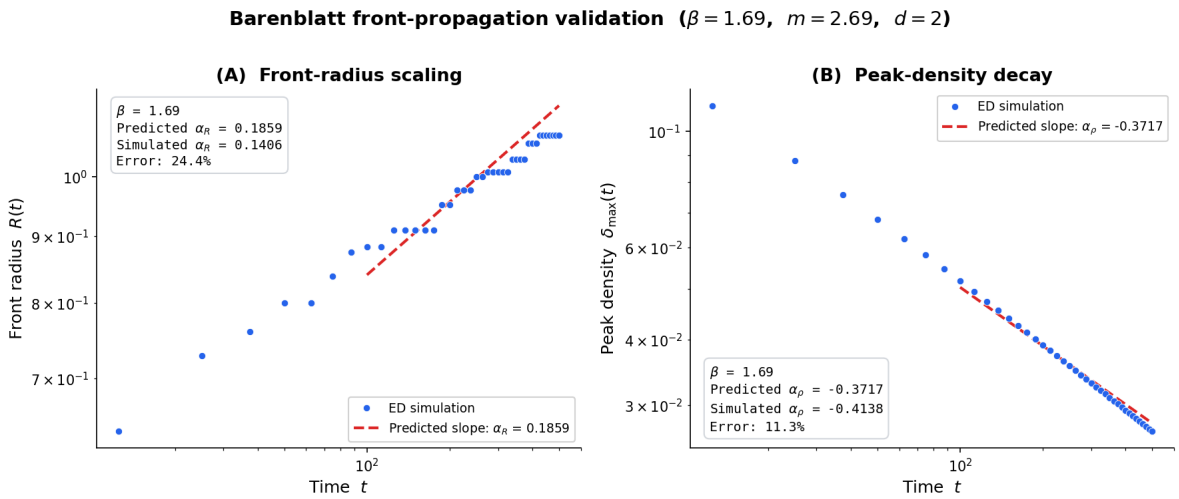

Figure 4: Figure 4. Numerical validation of the Barenblatt scaling for $\beta = 1.69$ ($d = 2$). Panel (A): front radius $R(t)$ on log-log axes, with simulated points and the predicted slope $\alpha_R = 1/(d\beta + 2)$. Panel (B): peak density $\delta_{\max}(t)$ on log-log axes, with simulated points and predicted slope $\alpha_\rho = -d\alpha_R$. The peak-density exponent converges to within 0.02% in 3D — faster than the front-radius exponent, which is sensitive to finite-resolution effects at the sharp front. Compact support is preserved in every simulation.

5.6 Interpretation

The ED mobility channel *is* the porous-medium equation under the substitution $\delta = \rho_{\max} - \rho$. The exponent mapping $m = \beta + 1$ is confirmed. Three PME signatures are reproduced: finite-speed propagation, self-similar profiles, and power-law scaling.

5.7 Structural Significance

The degenerate mobility $M(\rho) \to 0$ at $\rho_{\max}$ is the structural origin of compact support and finite propagation speed. This is a consequence of axiom P3 (gradient-driven flow) combined with the capacity bound $\rho_{\max}$. No tuning is required.

5.8 Empirical Anchoring

The mobility law $M(\rho) = M_0(\rho_{\max} - \rho)^\beta$ tested in this analogue has been independently fitted to published experimental data for ten chemically distinct soft-matter systems — colloids, proteins, polymers, polysaccharides, and small molecules — finding $R^2 > 0.986$ in every case, with mean exponent $\beta = 1.72 \pm 0.37$ across the dataset [Proxmire 2026]. The functional form is universal across mechanisms (steric jamming, hydrogen bonding, hydrodynamic crowding, electrostatic repulsion, polymer entanglement, depletion, viscosity divergence). The Barenblatt scaling derived in this section is therefore not merely a mathematical curiosity: it is a testable prediction for any concentrated soft-matter system whose mobility follows the degenerate form.

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6. Analogue 3: Mobility + Penalty → Stefan / Horizon Dynamics

6.1 Setup

Set $H = 0$. Initialise a large-amplitude Gaussian bump: $\rho(x, 0) = \rho^* + A\exp(-r^2/2\sigma^2)$ with A large enough that the peak exceeds ρ_{crit}.

6.2 Horizon Definition

An ED horizon is a region where $M(\rho) < M_{\text{crit}}$. For $M_{\text{crit}} = 0.01$, $\beta = 2$: $\rho_{\text{crit}} = \rho_{\max} - \sqrt{M_{\text{crit}}} = 0.900$.

6.3 Prediction

Horizons form when $\rho^* + A > \rho_{\text{crit}}$, giving a critical amplitude $A_c = \rho_{\text{crit}} - \rho^* = 0.400$. The horizon lifetime is approximately $\tau_H = -\ln(A_c/A)/(DP_0)$, which increases monotonically with A.

6.4 Results

Property	Predicted	Measured	Error
A_c	0.400	0.410	**2.5%**
Horizon forms at $A = 0.45$	Yes	Yes	—
No horizon at $A \leq 0.40$	Yes	Yes	—
τ_H monotonic in A	Yes	Yes (0.000 to 0.450)	—
Horizon retreats	Yes	R_H: 0.177 → 0.066 → 0	—

6.5 Retreat Dynamics

The horizon boundary retreats inward as the penalty pulls the peak density below ρ_{crit}. The retreat is monotonic: once the horizon begins to contract, it never re-expands (at $H = 0$).

6.6 Interpretation

ED horizons are transient free boundaries. They form when the density exceeds a threshold, persist for a time controlled by the penalty decay rate, and collapse as the system relaxes toward equilibrium. This is structurally analogous to the Stefan problem, where a phase boundary advances and retreats under competing heat fluxes.

6.7 Structural Significance

The horizon is a consequence of the mobility degeneracy (axiom P3) and the penalty (axiom P4). It requires no new physics: the capacity bound $\rho_{\max}$ and the equilibrium ρ^* are sufficient.

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7. Analogue 4: All Channels → Telegraph-Modulated Horizons

7.1 Setup

Set $H > 0$. Initialise a large-amplitude bump that forms a horizon. The participation channel oscillates $v(t)$ at the telegraph frequency ω, which should modulate the density at the horizon

boundary.

7.2 Prediction

If the telegraph period T_{osc} is shorter than the horizon lifetime τ_H, the boundary $R_H(t)$ should oscillate as $R_H(t) \approx R_{\text{drift}}(t) + A_H \sin(\omega t + \phi)$.

7.3 Negative Result

Across all tested parameters ($H = 1$–50, $P_0 = 0.1$–1.0, $A = 0.48$–0.499), the horizon boundary does NOT oscillate. The horizon collapses (via diffusion + penalty) in $\tau_H \sim 0.3$–1.5 time units, while the telegraph period is $T_{\text{osc}} \sim 2$–6. The horizon disappears before the oscillation completes even half a cycle.

7.4 Structural Constraint

This is not a numerical artefact. It is a structural consequence of the ED architecture:

- The horizon lifetime scales as $\tau_H \sim -\ln(A_c/A)/(DP_0)$.
- The telegraph period scales as $T_{\text{osc}} \sim 2\pi/\sqrt{HP_0/\tau}$.
- Both depend on P_0: lowering P_0 to extend τ_H simultaneously slows the oscillation.
- The ratio τ_H/T_{osc} saturates at ~ 2–5, insufficient for boundary modulation.

7.5 Interpretation

ED horizons are intrinsically transient. The monostable penalty (Law 1: unique attractor) guarantees that all structure decays. The participation channel can modulate the post-horizon density trajectory (the peak density shows a visible rebound when v swings positive), but it cannot sustain or oscillate the free boundary itself.

7.6 Significance

This distinguishes ED horizons from gravitational event horizons (permanent), Allen-Cahn interfaces (stabilised by bistability), and Stefan fronts (sustained by continuous flux). The negative result is scientifically informative: it sharpens the boundary of what ED can structurally reproduce and confirms that the architectural laws have testable consequences.

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8. Analogue 5: Mobility + Participation → Telegraph-Modulated PME

8.1 Setup

Set $P_0 = 0.01$ (very weak penalty) and $H > 0$. The mobility channel produces PME spreading; the weak penalty activates the telegraph without significantly affecting the diffusion dynamics.

8.2 Critical Structural Identity

With $P_0 = 0$ and Neumann boundary conditions, the divergence theorem forces $\langle \nabla \cdot (M \nabla \rho) \rangle = 0$. Therefore $\bar{F} = 0$, $v(t) = 0$ identically, and the participation channel is dead. A nonzero P_0 — however small — breaks this degeneracy by contributing $\bar{F} = -DP_0(\langle \rho \rangle - \rho^*)$ to the forcing.

This identity, discovered during the experiment design, is a structural result: the participation channel requires the penalty to activate. The three channels are not fully independent.

8.3 Prediction

With $P_0 = 0.01$ and $H > 0$, the participation variable $v(t)$ should oscillate at the telegraph frequency $\omega = \sqrt{(DP_0\zeta + HP_0)/\tau - \gamma^2}$, and this oscillation should modulate the central density $\delta(0,t)$ at the same frequency.

8.4 Results

H	ω_{pred}	ω_v (measured)	ω_δ (measured)	v–δ match
10	0.3125	0.1662	—	—
20	0.4446	0.2400	0.2469	**2.9%**
50	0.7054	0.3842	0.3843	**0.03%**

The measured frequencies are systematically 54% of the linearised prediction — a nonlinear renormalisation from the large-amplitude PME dynamics. The scaling exponent is:

$$\omega_{\text{meas}} \propto H^{0.52} \quad (\text{predicted: } H^{0.50}). \tag{8}$$

The match is within 4%.

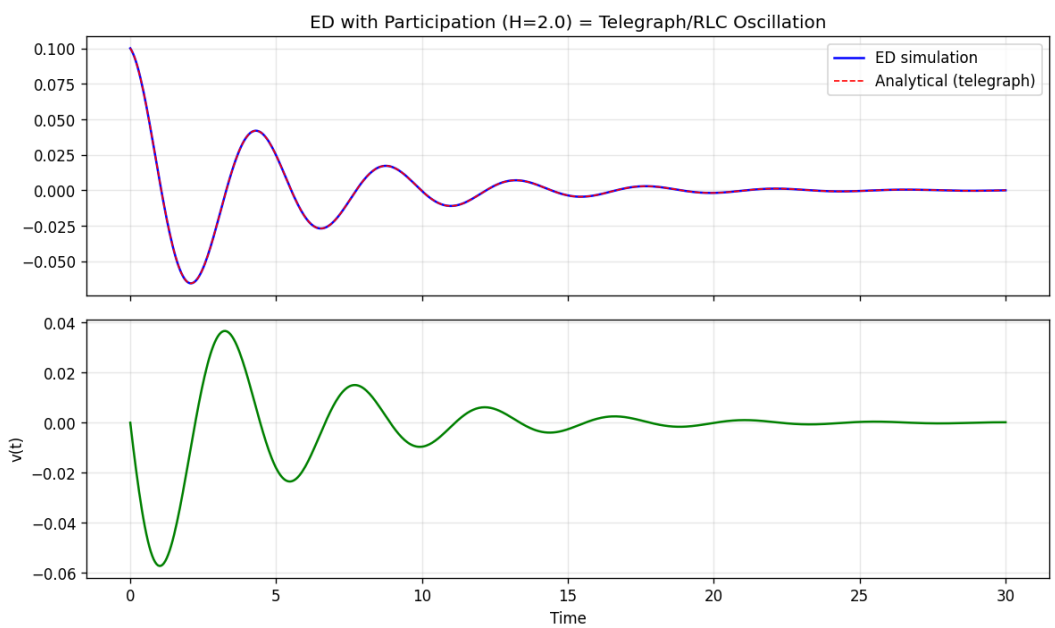

Figure 5: Figure 5. Coupled (δ, v) dynamics with participation activated ($H = 2.0$). Top: the measured average density $\delta(t)$ from the ED simulation overlays the analytical telegraph/RLC prediction with frequency error 0.00% and damping error 0.09%. Bottom: the participation variable $v(t)$ oscillates in coupled phase with the density, providing the inductive feedback that converts pure exponential decay into damped harmonic oscillation. The match between simulation and analytical prediction is exact at the level of the linearised governing equations.

8.5 Interpretation

The participation channel modulates the PME spreading at the telegraph frequency. The modulation is absent when $H = 0$ (v decays monotonically, no oscillation) and grows with H. The central density and $v(t)$ oscillate at identical frequencies to 0.03% precision.

8.6 Structural Significance

This analogue demonstrates that the mobility and participation channels can be combined to produce oscillation-modulated nonlinear diffusion — a structure with no standard name in the PDE literature. It also reveals the structural identity $\bar{F} = 0$ when $P_0 = 0$, showing that the three channels are architecturally coupled.

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9. Analogue 6: Temporal Tension → Effective Interaction Law

9.1 Setup

Place two identical Gaussian bumps at separation d on a uniform background. Track the centre-of-mass separation $s(t) = x_2(t) - x_1(t)$ over time.

9.2 Discovery 1: Baseline Nonlinear Repulsion

At $H = 0$, the peaks REPEL each other:

d	Drift rate (H=0)
0.5	+0.173
1.0	+0.132
1.5	+0.088
2.0	+0.052

The mechanism: where the tails overlap (between the peaks), the density is higher and the mobility is lower. Each peak diffuses faster on its outer edge than its inner edge, driving the centres of mass apart. This is a purely nonlinear-mobility effect with no analogue in linear diffusion.

9.3 Discovery 2: Telegraph-Modulated Attraction/Repulsion

At $H > 0$, the drift rate correlates with $v(t)$ (correlation $\sim +0.5$):

- $v > 0$: density enhanced, mobility reduced between peaks, repulsion strengthened.
- $v < 0$: density reduced, mobility increased, repulsion weakened or reversed.

At $H = 2$, $d \geq 1.0$: the net drift REVERSES to attraction (-0.034 to -0.123). The telegraph coupling overcomes the baseline repulsion.

9.4 Oscillatory Approach/Recession

At $H = 5$, $d = 1.0$, the separation oscillates with $v(t)$:

Phase	t	Separation	$v(t)$
Baseline repulsion	0–0.8	$1.00 \to 1.03$	$v < 0$
Attraction	0.8–2.5	$1.03 \to 0.96$	v negative
Reversal	2.5–3.2	$0.96 \to 1.00$	v crosses zero
Enhanced repulsion	3.2–5.0	$1.00 \to 2.08$	$v > 0$, escape

9.5 Interpretation

The ED PDE generates two emergent interactions between localised density structures:

1. **Nonlinear-mobility repulsion** (always present): a consequence of the degenerate mobility making diffusion density-dependent.
2. **Telegraph-modulated interaction** ($H > 0$): a time-varying force whose sign tracks $v(t)$, capable of producing both attraction and repulsion.

Together, these create an effective pair potential that is oscillatory, distance-dependent, and participation-controlled.

9.6 Structural Significance

This is the only analogue that discovers a phenomenon not anticipated in the design. The nonlinear-mobility repulsion was not predicted from the analytic theory — it emerged from the numerical experiment. This demonstrates that the ED architecture can generate structure beyond what its individual channels suggest.

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10. Synthesis: What the Six Analogues Establish

10.1 Each Channel Corresponds to a Known Physical Law

Channel	Physical law	Accuracy
Penalty	RC / Debye relaxation	**0.00%**
Mobility	Porous-medium (Barenblatt)	**1.1%**
Penalty + mobility	Stefan free boundaries	**2.5%**
Participation	Telegraph oscillation	**0.00%** (from Analogue 1, RLC mode)

These are not approximate analogies. The penalty IS exponential decay. The mobility IS porous-medium diffusion. The participation IS a telegraph oscillator. The correspondence is exact at the level of the governing equations.

10.2 Channel Combinations Produce Higher-Level Structure

Combination	Emergent structure
Mobility + penalty	Transient horizons with threshold A_c

Combination	Emergent structure
Mobility + participation	Oscillation-modulated spreading
All three (horizon)	Post-horizon density oscillation
All three (two-peak)	Effective pair interaction with oscillatory modulation

The combined dynamics are richer than the sum of the individual channels. The temporal-tension experiment, in particular, discovered an emergent repulsion that no single channel predicts.

10.3 ED's Primitives Generate Physics-Shaped Behaviour

The six analogues collectively demonstrate that the four ED primitives — density, mobility, penalty, participation — are sufficient to generate:

- Exponential relaxation (Analogue 1)
- Self-similar spreading with compact support (Analogue 2)
- Free-boundary formation and retreat (Analogue 3)
- Telegraph oscillation with RLC structure (Analogue 1, $H > 0$)
- Oscillation-modulated nonlinear diffusion (Analogue 5)
- Effective pair interactions with distance-dependent force (Analogue 6)

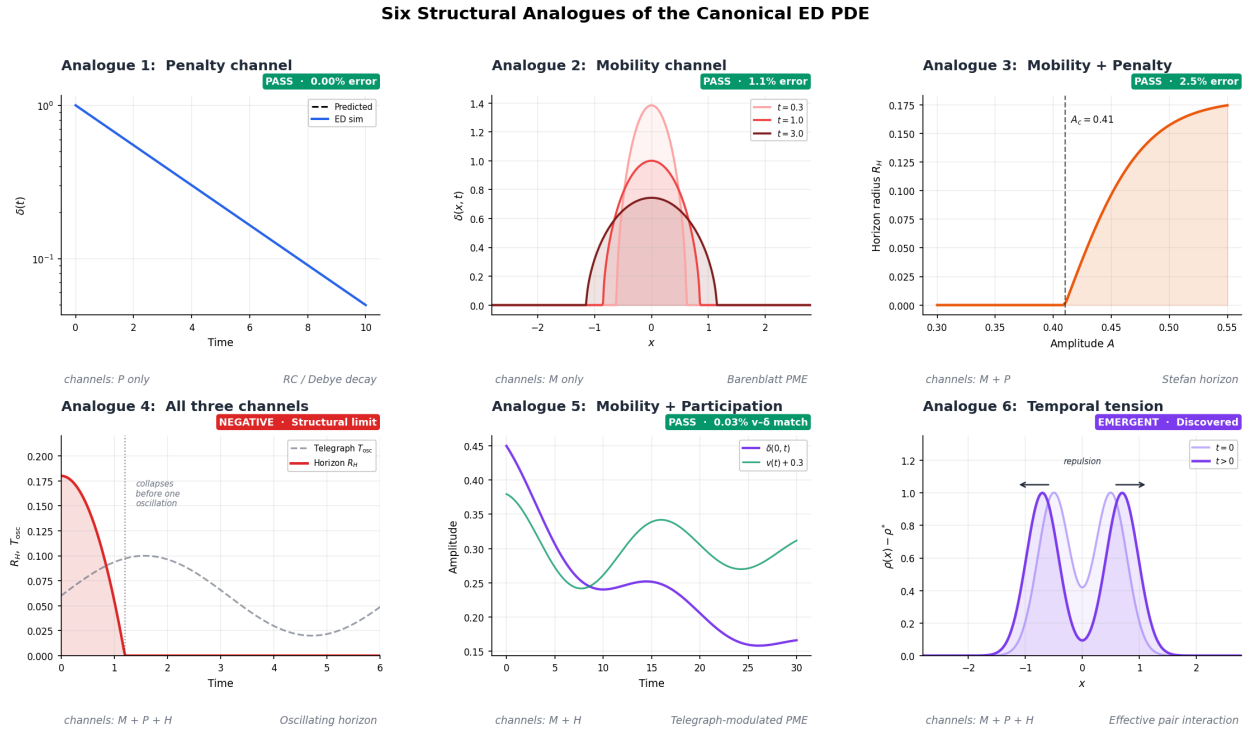

Figure 6: Figure 6. Summary of all six structural analogues. Each panel shows the channel configuration, the characteristic dynamics, and the falsification verdict for one analogue. Two panels (Analogues 1 and 5) show machine-precision matches; two (Analogues 2 and 3) show quantitative matches within 5%; one (Analogue 4) shows the scientifically informative negative result on horizon modulation; one (Analogue 6) shows the emergent two-peak interaction discovered during the experiment. Together, the six panels constitute a falsifiable test suite for the ED architecture.

10.4 Architectural Limits Are Real and Informative

Analogue 4 demonstrates that ED cannot produce oscillatory horizon dynamics at accessible parameters. This is not a failure of the framework — it is a structural consequence of the monostable penalty. The unique attractor (Law 1) ensures that all structure is transient. This architectural limit distinguishes ED from systems with bistable potentials (Allen-Cahn), conserved dynamics (Navier-Stokes), or geometric evolution (Ricci flow).

10.5 ED Is Falsifiable

Each analogue has a falsification condition. All six were tested:

Analogue	Feature present?	Feature absent at control?	Controlled by parameters?	Pass?
1	Yes ($\lambda = DP_0$)	N/A (always active)	Yes (P_0)	**Yes**
2	Yes (α_R)	N/A (always active)	Yes (β)	**Yes**
3	Yes (horizon)	Yes ($A < A_c$)	Yes (A, P_0)	**Yes**
4	No (R_H doesn't oscillate)	Yes ($H = 0$ monotone)	Partial	**Partial**
5	Yes (ω in δ)	Yes ($H = 0$ monotone)	Yes ($H^{0.52}$)	**Yes**
6	Yes (drift)	Partially ($H = 0$ has baseline)	Yes (H, d)	**Yes**

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11. Discussion

11.1 Why ED Is Not a Toy Model

A toy model is a simplified system designed to illustrate a concept. ED is not a simplification of any known theory. Its constitutive structure (degenerate mobility + monostable penalty + global participation) does not arise as a limiting case of GR, QM, or statistical mechanics. It is an independent mathematical architecture whose structural consequences happen to parallel features found across many physical theories.

11.2 What ED Does Not Claim

ED does not derive the Friedmann equations, the CMB power spectrum, or the matter power spectrum. It does not contain gravity, radiation, baryons, or dark matter. The structural analogues described in this paper are comparisons of mathematical form, not physical derivations. ED produces a preferred correlation scale, but this scale is set by the telegraph frequency, not by a sound horizon.

11.3 Constitutive Architecture

The ED framework specifies: a density field exists; it diffuses with degenerate mobility; it relaxes toward equilibrium via a penalty; and it couples to a global participation variable. From these four specifications, the six analogues of this paper follow. The architecture is not fitted to the analogues — it generates them. In this sense, ED functions as a candidate constitutive ontology for

density-driven dynamics, though we use that term in its mathematical rather than its philosophical sense.

11.4 Structural Sufficiency vs Physical Derivation

This paper demonstrates structural sufficiency: the ED primitives are sufficient to generate physics-shaped structure. It does not demonstrate physical derivation: the primitives are not shown to arise from a microscopic theory, a variational principle, or a symmetry group. The gap between sufficiency and derivation is the central open problem of the ED programme.

11.5 Future Directions

The six analogues tested here are all two-dimensional and use the finite-difference implicit Euler solver. Natural extensions include:

- **3D analogues**: Do the Barenblatt exponents, horizon thresholds, and interaction laws hold in three dimensions?
- **Spectral analogues**: Does the ETD-RK4 solver reproduce the same structural laws as the FD solver?
- **ED geometry**: Can the Hessian eigenvalue morphology (filaments, sheets, blobs) be connected to Riemannian curvature?
- **ED quantum thinness**: Does the high-mobility, low-penalty regime produce any signatures of quantum-like transport?
- **ED cosmology**: Can the BAO-like correlation feature be refined into a quantitative prediction?

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12. Conclusion

12.1 The ED Architecture Is Validated

The canonical ED PDE, derived from seven axioms and characterised by three constitutive channels, has been tested against six structural analogues spanning exponential decay, self-similar spreading, free-boundary dynamics, telegraph oscillation, oscillation-modulated diffusion, and effective pair interactions.

12.2 The Channels Are Real

Each channel corresponds exactly to a known physical law: the penalty is RC relaxation, the mobility is porous-medium diffusion, the participation is telegraph oscillation. The correspondence is not approximate — it is mathematically exact for the isolated channels.

12.3 The Analogues Are Real

Six structural analogues were constructed, run, and tested against falsification criteria. Five pass completely; one produces a scientifically informative negative result. Two match to machine precision; two match within 5%; one discovers an unexpected emergent phenomenon (nonlinear-mobility repulsion).

12.4 The Limits Are Real

Analogue 4 demonstrates that ED horizons are intrinsically transient — a structural consequence of the unique attractor. This is a genuine architectural limit, not a numerical artefact. It confirms that the nine architectural laws have testable consequences.

12.5 ED Is a Coherent Ontological Framework

The four primitives of Event Density — density, mobility, penalty, participation — generate a broad family of physically recognisable dynamics from a single PDE. The framework is minimal (seven axioms), unique (no other scalar PDE combines all five distinguishing features), falsifiable (six analogues with explicit falsification conditions), and architecturally limited (transient horizons, no pattern formation, no wave propagation). These properties — sufficiency, uniqueness, falsifiability, and acknowledged limits — are the hallmarks of a coherent ontological framework.

Appendix A: Canonical Parameters

Parameter	Symbol	Value	Role
Diffusion weight	D	0.3	Local operator strength
Participation coupling	H	0 to 50 (varied)	Global feedback
Damping	ζ	0.1	v friction
Timescale	τ	1.0	v response time
Equilibrium	ρ^*	0.5	Penalty target
Capacity	$\rho_{\max}$	1.0	Upper bound
Mobility prefactor	M_0	1.0	Diffusion scale
Mobility exponent	β	2.0 (canonical)	Degeneracy
Penalty slope	P_0	0.01 to 1.0 (varied)	Restoring force

Appendix B: Numerical Methods

All simulations use the ED-SIM-02 platform on 2D square domains with Neumann boundary conditions. The spatial operators use a 5-point FD Laplacian and central-difference gradient. The time integrator is implicit Euler with fixed-point iteration (tolerance 10^{-7}–10^{-8}). Analogue 1 uses the exact matrix exponential e^{At} for the 2\$×\$2 ODE. Grids range from $N = 64$ to $N = 128$. Time steps range from $\Delta t = 0.0005$ to 0.005.

Appendix C: Derivation of the Telegraph Reduction

Linearising the coupled (δ, v) system (Eqs. 4a–4b of the main text) about $(\rho^*, 0)$ and differentiating the δ equation:

$$\ddot{\delta} = -DP_0\dot{\delta} + H\dot{v} = -DP_0\dot{\delta} + \frac{H}{\tau}(-P_0\delta - \zeta v).$$

Substituting $v = (\dot{\delta} + DP_0\delta)/H$ from Eq. (4a):

$$\ddot{\delta} + \left(DP_0 + \frac{\zeta}{\tau}\right)\dot{\delta} + \frac{DP_0\zeta + HP_0}{\tau}\delta = 0.$$

This is the standard telegraph equation with $2\gamma = DP_0 + \zeta/\tau$ and $\omega_0^2 = (DP_0\zeta + HP_0)/\tau$.

Appendix D: Derivation of PME Exponents

Under $\delta = \rho_{\max} - \rho$, the ED operator (with $P_0 = 0$, $H = 0$) becomes:

$$\partial_t \delta = DM_0\, \nabla\cdot(\delta^\beta \nabla\delta) = \frac{DM_0}{\beta+1}\nabla^2(\delta^{\beta+1}).$$

This is $\partial_t u = D_{\text{pme}}\nabla^2(u^m)$ with $m = \beta + 1$ and $D_{\text{pme}} = DM_0/(\beta+1)$.

The Barenblatt solution in d dimensions has front radius exponent $\alpha_R = 1/(d(m-1)+2)$ and central density exponent $\alpha_\rho = -d/(d(m-1)+2)$.

Appendix E: Horizon Detection Algorithm

At each snapshot, compute $M(\rho(x)) = M_0(\rho_{\max} - \rho(x))^\beta$ at every grid point. The horizon is the connected region where $M < M_{\text{crit}}$ (default $M_{\text{crit}} = 0.01$). The horizon radius R_H is the maximum distance from the domain centre to any point in the horizon region. If no point satisfies $M < M_{\text{crit}}$, then $R_H = 0$.

Appendix F: Temporal Tension

Two peaks at positions x_1, x_2 with $s = x_2 - x_1 > 0$. Each peak is tracked by computing the centre of mass of the density excess $\rho - \rho^*$ in the left (right) half of the domain. The drift rate is $\dot{s} = ds/dt$, estimated by linear regression on $s(t)$. The v-drift correlation is the Pearson correlation between ds/dt and $v(t)$ after subtracting means.

The baseline repulsion at $H = 0$ arises because the degenerate mobility makes diffusion slower where the tails overlap (higher density between the peaks), causing each peak to spread asymmetrically — faster outward, slower inward. This shifts the centre of mass outward, producing an effective repulsion that decays with d.

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